NAMIBIA UNIVERSITY OF SCIENCE AND TECHNOLOGY

Faculty of Computing and Informatics

School of Computing

Department of Software Engineering

QUALIFICATION: BACHELOR OF COMPUTER SCIENCE	
QUALIFICATION CODE: 07BACS	LEVEL: 7
COURSE: ARTIFICIAL INTELLIGENCE	COURSE CODE: ARI711S
SESSION: MAY 2023	PAPER: THEORY
DURATION: 2 HOURS	MARKS: 50

SECOND-OPPORTUNITY TEST	
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This paper consists of **5 pages** (including this front page)

INSTRUCTIONS

1. This paper contains 4 questions.
2. Answer all questions on the question paper.
3. Marks/scores are indicated at the right end of each question.
4. Calculators are permitted.
5. NUST examination rules and regulations apply.

NAME & SURNAME									
STUDENT NUMBER									

FOR STAFF USE ONLY	
QUESTION 1: SEARCH	
QUESTION 2: ADVERSARIAL SEARCH	
QUESTION 3: CSP	
QUESTION 4: MDP	
TOTAL MARK	
PERCENTAGE	

QUESTION 1: SEARCH **[6]**

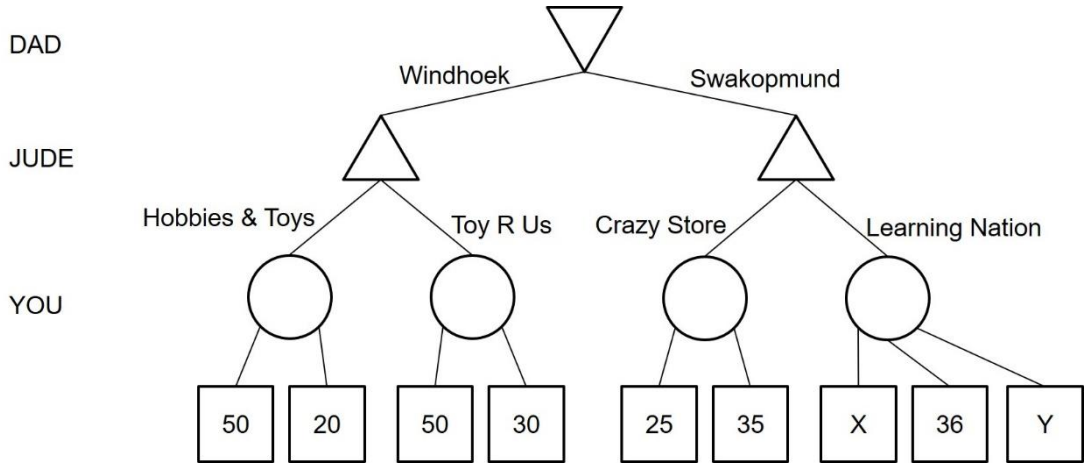
Select whether the following statements are either TRUE or FALSE. Cross [X] the corresponding box for each question, to indicate your answer.

Assume we are running $A*$ graph search with a consistent heuristic h. Assume the optimal cost path to reach a goal has a cost $c*$. Then we have that:

		True	False	
1.1	All nodes *n* reachable from the start state satisfying $g(n) < c*$ will be expanded during the search.			(2)
1.2	All nodes *n* reachable from the start state satisfying $f(n) = g(n) + h(n) < c*$ will be expanded during the search.			(2)
1.3	All nodes *n* reachable from the start state satisfying $h(n) < c*$ will be expanded during the search.			(2)

QUESTION 2: ADVERSARIAL SEARCH - EXPECTIMAX **[10]**

Your little brother Jude, has a birthday and he was promised a toy. However, Jude has been misbehaving lately and Dad thinks he deserves the least expensive present. Jude, of course, wants the most expensive toy. Dad will pick the city from which to buy the toy, Jude will pick the store and you get to pick the toy itself. You don't want to take sides so you decide to pick a toy at random. All prices (including X and Y) are assumed to be nonnegative.



2.1 Fill in the values of all the nodes that don't depend on X or Y. (4)

2.2 What values of X will make Dad pick Windhoek regardless of the price of Y? (3)

2.3 We know that Y is at most $30. What values of X will result in a toy from Crazy Store regardless of the exact price of Y? (3)

QUESTION 3: CSP **[14]**

After years of struggling through mazes, Pacman has nally made peace with the ghosts, Blinky, Pinky, Inky, and Clyde, and invited them to live with him and Ms. Pacman. The move has forced Pacman to change the rooming assignments in his house, which has 6 rooms. He has decided to gure out the new assignments with a CSP in which the variables are Pacman (**P**), Ms. Pacman (**M**), Blinky (**B**), Pinky (**K**), Inky (**I**), and Clyde (**C**), the values are which room they will stay in, from 1-6, and the constraints are:

i) No two agents can stay in the same room
ii) **P** > 3
iii) **K** is less than **P**
iv) **M** is either 5 or 6
v) **P** > **M**
vi) **B** is even
vii) **I** is not 1 or 6
viii) (**I-C**) = 1
ix) (**P-B**) = 2

3.1 **Unary constraints:** On the grid below cross out the values from each domain that are eliminated by enforcing unary constraints. (4)

P	1	2	3	4	5	6
B	1	2	3	4	5	6
C	1	2	3	4	5	6
K	1	2	3	4	5	6
I	1	2	3	4	5	6
M	1	2	3	4	5	6

3.2 **MRV:** According to the Minimum Remaining Value (MRV) heuristic, which variable should be assigned to first? (2)

○ P ○ B ○ C ○ K ○ I ○ M

3.3 **Forward Checking**: For the purposes of decoupling this problem from your solution to the previous problem, assume we choose to assign P first, and assign it the value 6. What are the resulting domains after enforcing unary constraints (from part 3.1) and running forward checking for this assignment? (4)

P						6
B	1	2	3	4	5	6
C	1	2	3	4	5	6
K	1	2	3	4	5	6
I	1	2	3	4	5	6
M	1	2	3	4	5	6

3.4 **Iterative Improvement:** Instead of running backtracking search, you decide to start over and run iterative improvement with the min-conflicts heuristic for value selection. Starting with the following assignment: P:6, B:4, C:3, K:2, I:1, M:5 (4)

First, for each variable write down how many constraints it violates in the table below. Then, in the table on the right, for all variables that could be selected for assignment, put an [X] in any box that corresponds to a possible value that could be assigned to that variable according to min-conflicts. When marking next values a variable could take on, only mark values different from the current one.

Variable	# violated
P	
B	
C	
K	
I	
M	

	1	2	3	4	5	6
P						
B						
C						
K						
I						
M						

QUESTION 4: MDP **[20]**

The following problems take place in various scenarios of the gridworld MDP (as in Assignment P3). In all cases, A is the start state and double-rectangle states are exit states.

From an exit state, the only action available is *Exit*, which results in the listed reward and ends the game (by moving into a terminal state X, not shown).

From non-exit states, the agent can choose either *Left* or *Right* actions, which move the agent in the corresponding direction. There are no living rewards; the only non-zero rewards come from exiting the grid.

Throughout this problem, assume that value iteration begins with initial values $V_0(s) = 0$ for all states s.

First, consider the following mini-grid. For now, the discount is $\gamma = 1$ and legal movement actions will always succeed (and so the state transition function is deterministic).

4.1 What is the optimal value $V^*(A)$? (1)

4.2 When running value iteration, remember that we start with $V_0(s) = 0$ for all s. What is the first iteration k for which $V_k(A)$ will be non-zero? (2)

4.3 What will $V_k(A)$ be when it is first non-zero? (1)

4.4 After how many iterations k will we have $V_k(A) = V^*(A)$? If they will never become equal, write *never*. (2)

Now the situation is as before, but the discount is less than 1.

4.5 If $\Upsilon = 0.5$, what is the optimal value $V^*(A)$? (3)

4.6 For what range of values Υ of the discount will it be optimal to go Right from A? Remember that $0 \leq \Upsilon \leq 1$. Write *all* or *none* if all or no legal values of have this property. (4)

Let's kick it up a notch! The *Left* and *Right* movement actions are now *stochastic* and *fail* with probability *f*. When an action fails, the agent moves up or down with probability f/2 each. When there is no square to move *up* or *down* into (as in the one-dimensional case), the agent stays in place. The *Exit* action does not fail.

For the following mini-grid, the failure probability is $f = 0.5$ and the discount is $\Upsilon = 1$

4.7 What is the optimal value $V^*(A)$? (1)

4.8 When running value iteration, what is the smallest value of k for which $V_k(A)$ will be non-zero? (1)

4.9 What will $V_k(A)$ be when it is first non-zero? (2)

4.10 After how many iterations *k* will we have $V_k(A) = V^*(A)$? If they will never become equal, write *never*. Motivate your answer. (3)

END OF TEST PAPER